Terrain models: tiles, hillshade, local relief

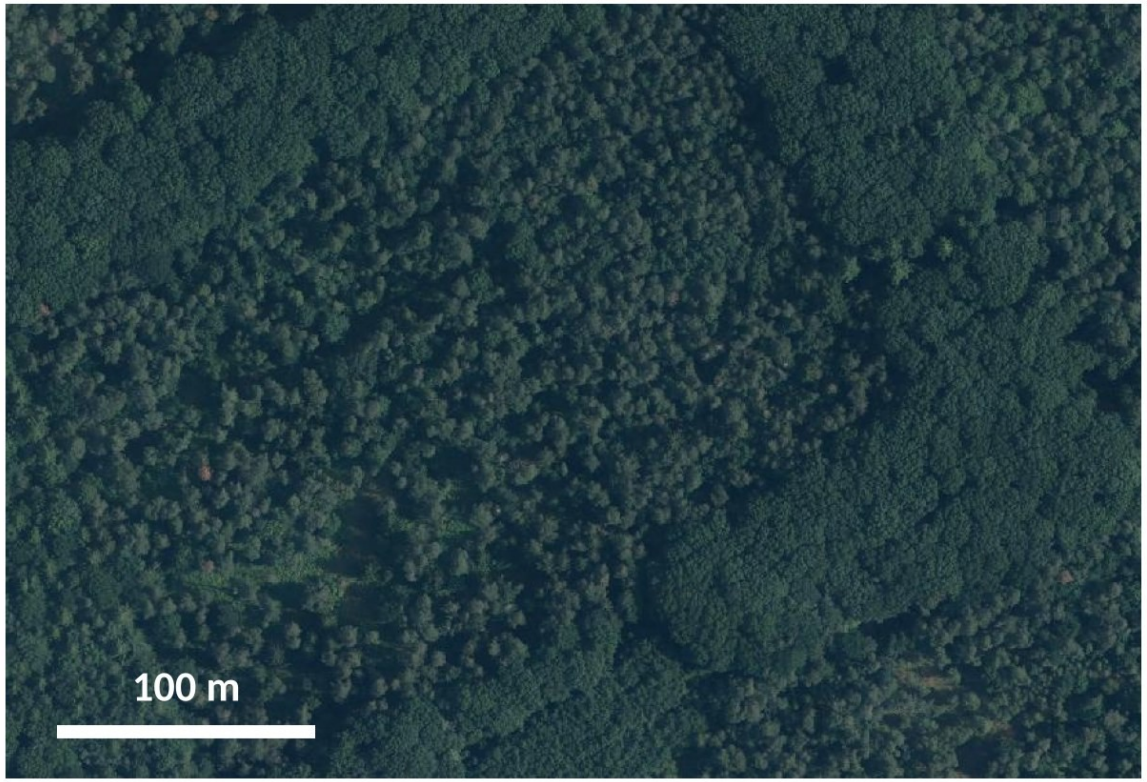
NRW DGM1 tiles in QGIS and Python · map algebra on the terrain

Case study: the burial mounds of the Uedemer Hochwald

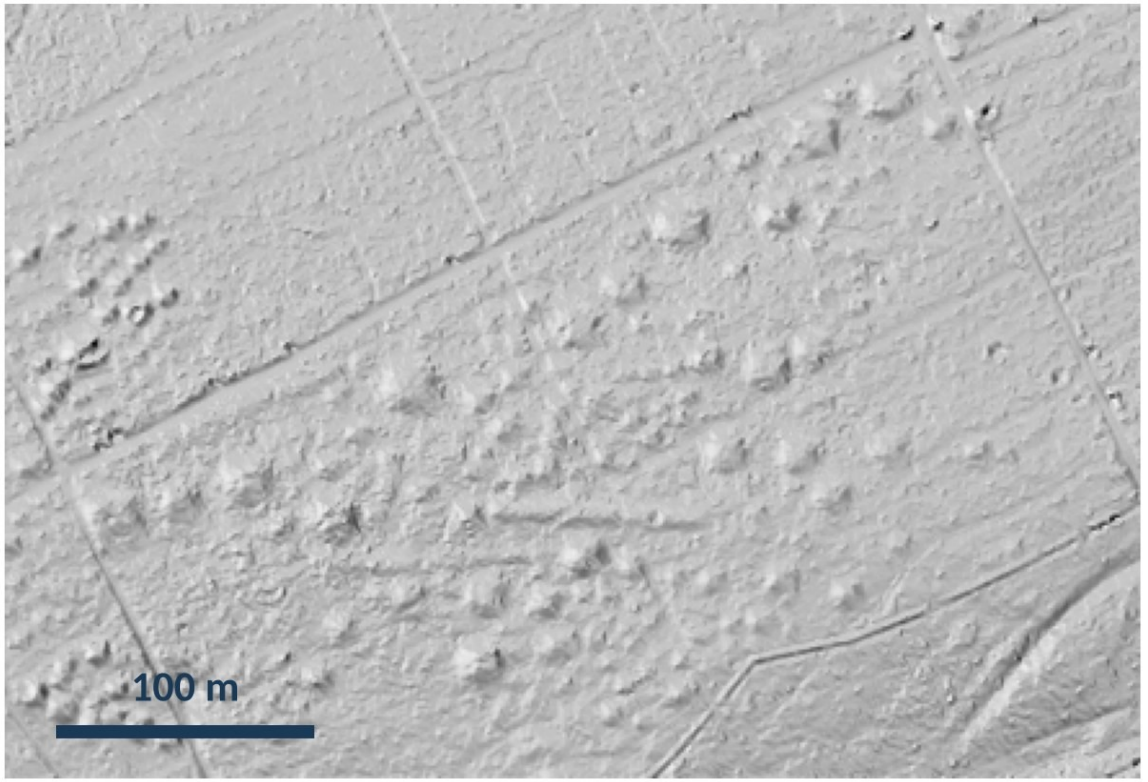


What the forest hides

orthophoto (DOP): trees



terrain model (DGM1), hillshade: mounds



Pause: How can an aircraft measure the ground under the trees?

DOP, DGM1, DOM1: © Geobasis NRW, dl-de/zero-2-0

Today

- **Terrain models:** laser scanning, surface vs terrain
- **Tiles:** find, download and combine 1 km tiles (virtual raster)
- **Hillshade:** light and shadow make 1 m visible
- **Local relief:** terrain − smoothed terrain (map algebra)
- **Mounds:** map, terrain model and algorithm compared



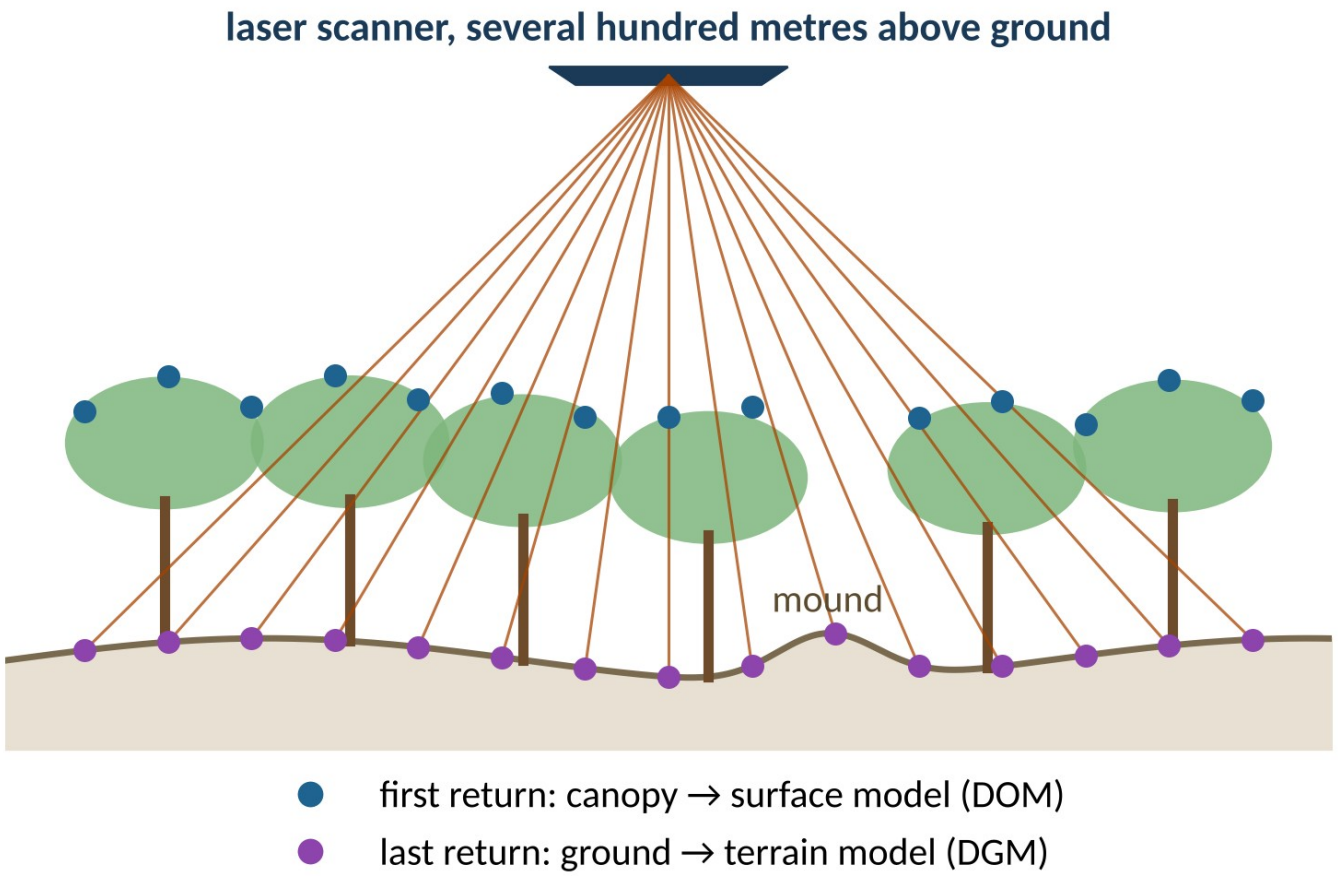
1 Terrain models

Laser scanning, surface and terrain

Terrain models | Tiles | Hillshade | Local relief | Mounds



Laser scanning from the air

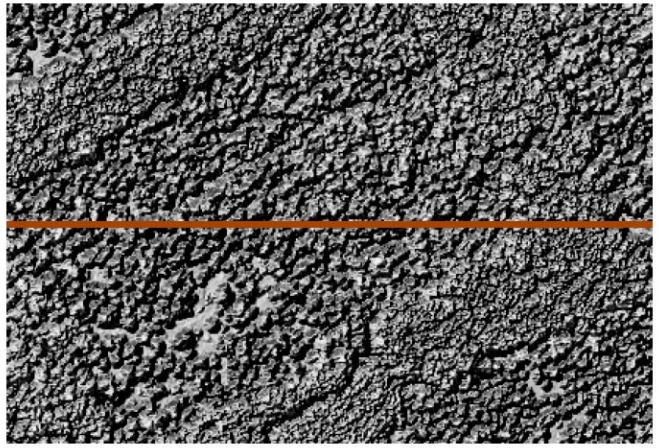


- Pulse travel time → range
- GNSS + inertial unit → position of every echo
- **First return:** canopy, roofs
- **Last return:** ground
- Ground points → terrain model

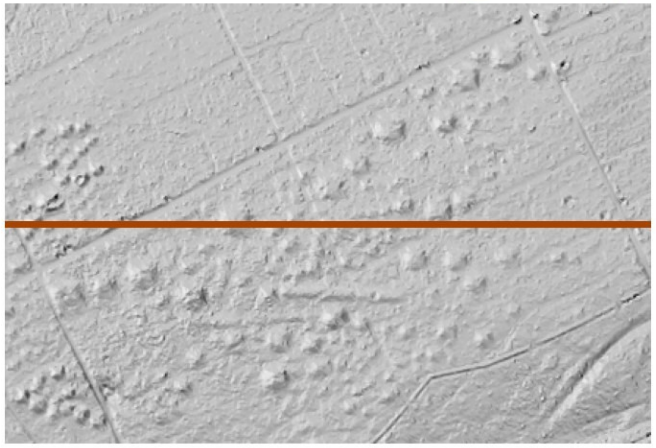
Wikipedia: Lidar · Airborne laser scanning (de)

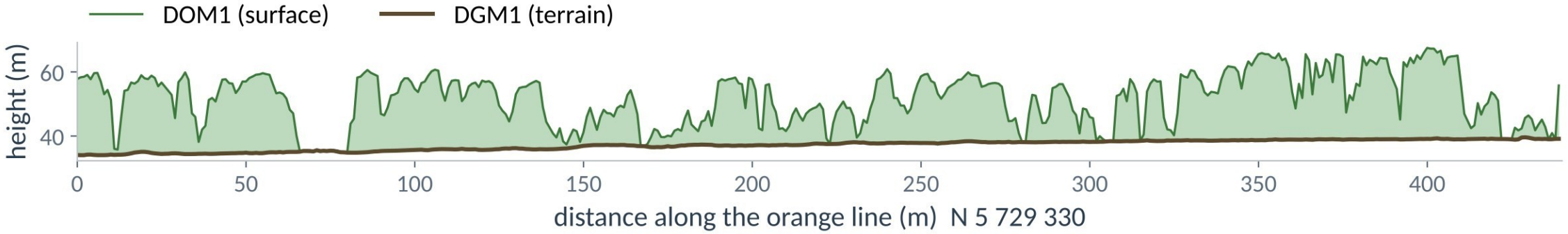
Surface model or terrain model?

surface model DOM1 (first returns)



terrain model DGM1 (ground)





DOM – DGM = height of trees and buildings (normalised surface model)

Wikipedia: Digital elevation model · Digitales Geländemodell (de)

DGM1, DOM1: © Geobasis NRW, dl-de/zero-2-0

The NRW terrain model DGM1

	DGM1 NRW (opengeodata.nrw.de, 2026-10)
grid	1 m × 1 m, heights in m above NHN (DHHN2016)
CRS	ETRS89 / UTM zone 32N, EPSG:25832
tiles	1 km × 1 km = 1000 × 1000 cells, GeoTIFF float32
coverage	35 860 tiles, 78.8 GB
survey years	2020–2025, different per tile
licence	dl-de/zero-2-0: free, name the source

Geobasis NRW, DGM1 file list https://www.opengeodata.nrw.de/produkte/geobasis/hm/dgm1_tiff/dgm1_tiff/



2 Tiles

Find, download, combine

Terrain models | **Tiles** | Hillshade | Local relief | Mounds



Six tiles of 1 km × 1 km



File name = bounding box

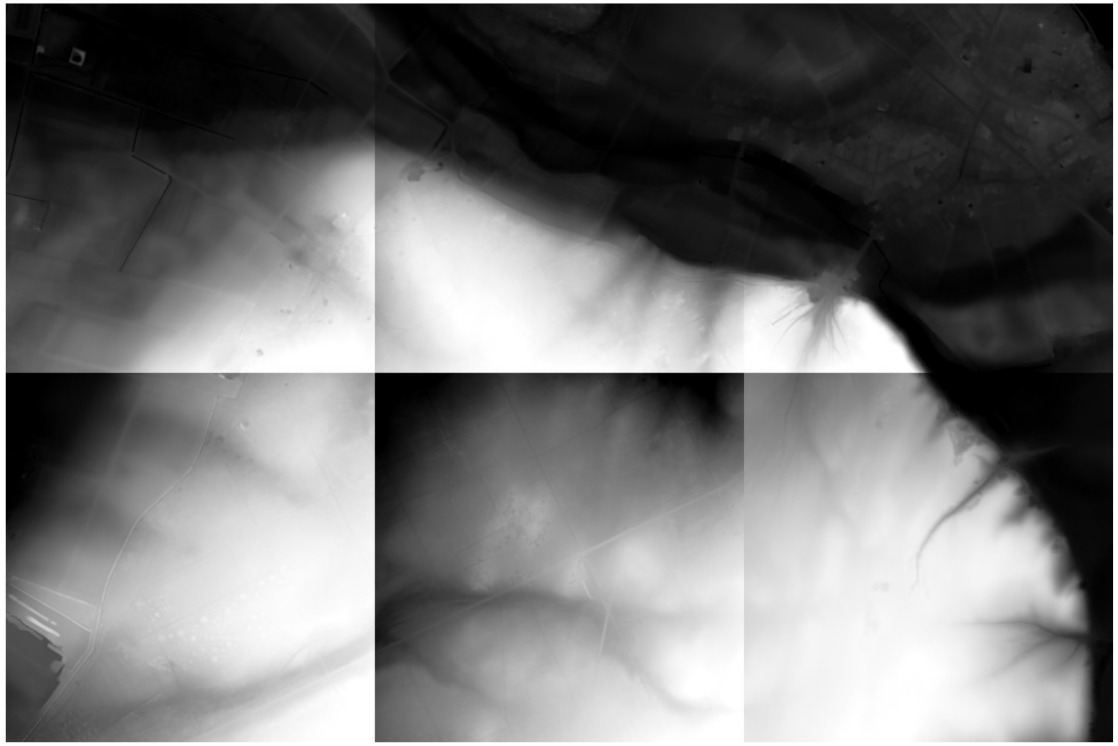
```
dgm1_32_316_5729_1_nw_2025.tif
```

- **32:** UTM zone (EPSG:25832)
- **316, 5729:** lower left corner in km
- **1:** tile size 1 km
- **2025:** survey year: only the server knows it

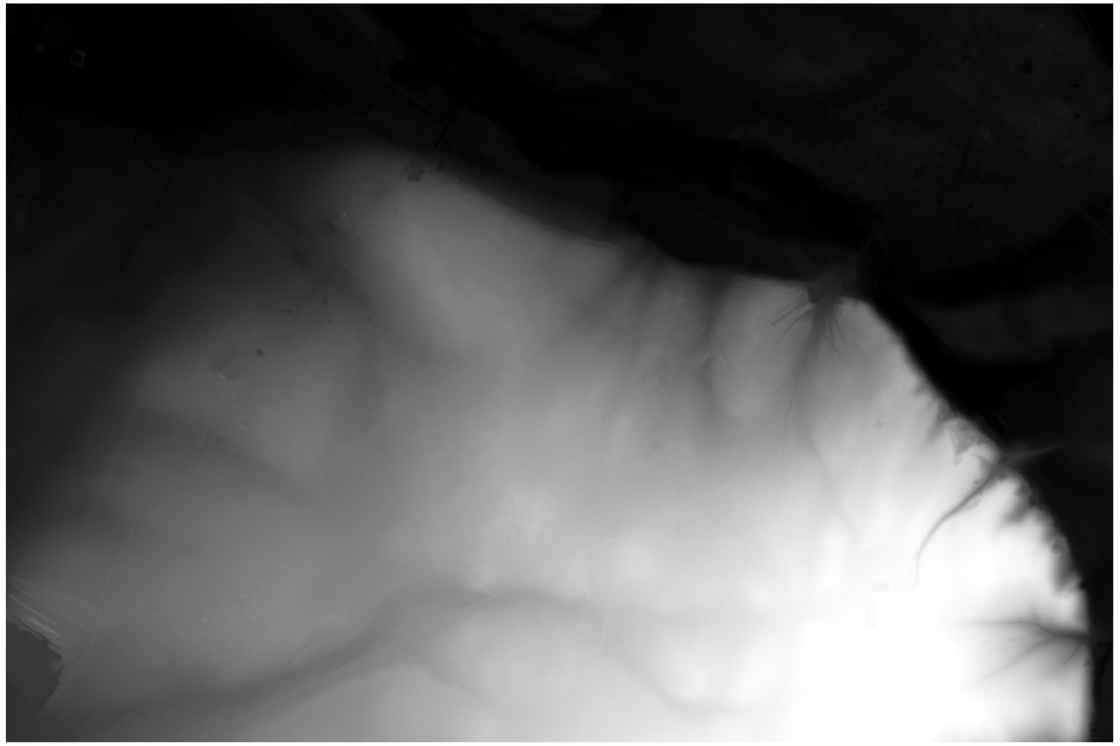
DGM1, DOM1: © Geobasis NRW, dl-de/zero-2-0

Six layers, six stretches

six layers: each tile its own min/max



one virtual raster: one stretch (18–65 m)



Pause: Are the seams in the data or in the display?

DGM1, DOM1: © Geobasis NRW, dl-de/zero-2-0

One layer from six files: the virtual raster

- A VRT is a few kB of XML
- It points to the tiles, copies no pixel
- One layer → one stretch, one hillshade, one Raster Calculator input
- **Merge:** a new 24 MB GeoTIFF, if a program needs one file

```
from glob import glob
from osgeo import gdal

tiles = glob("dgm1_uedemer*/*.tif")
# a VRT: pointers to the tiles
gdal.BuildVRT("uedem.vrt", tiles)

# or one array in memory:
from rasterio.merge import merge
arr, transform = merge(tiles)
```

Wikipedia: Tiled web map · GDAL



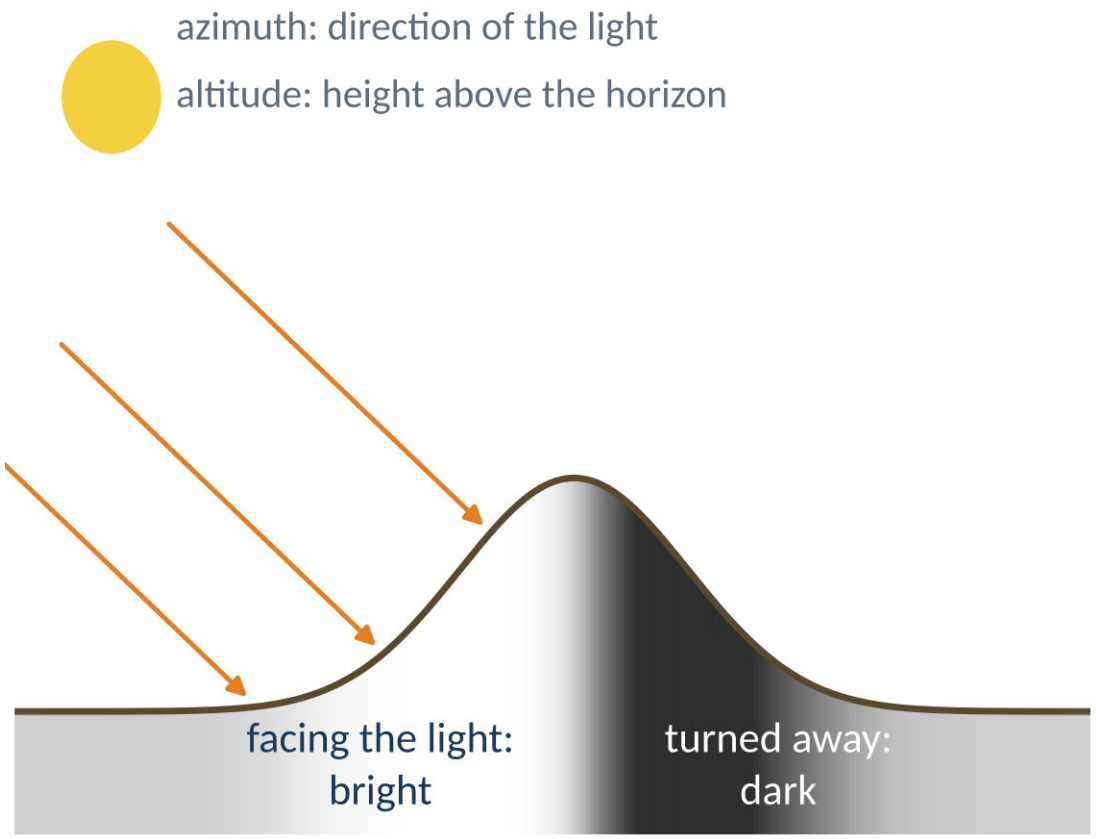
3 Hillshade

Light and shadow make one metre visible

Terrain models | Tiles | **Hillshade** | Local relief | Mounds



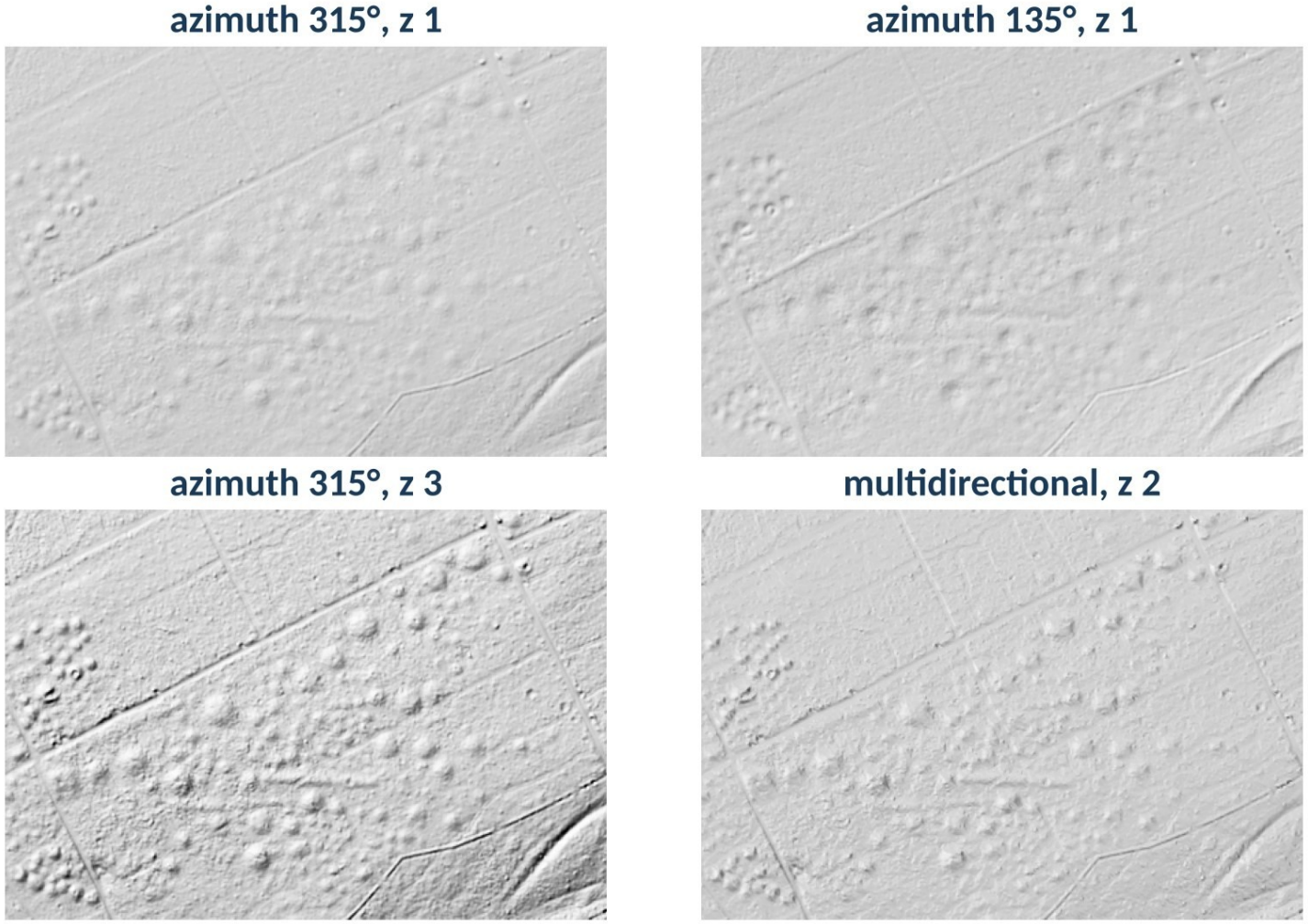
Hillshade: light from one direction



- Brightness = cos(angle between the surface and the light)
- Needs slope and aspect of each cell: a 3 × 3 window (focal)
- Default: light from the north-west (315°), 45° high
- **A grey value is not a height**

Wikipedia: Terrain cartography · Hillshading (de)

Azimuth and z factor



- **Azimuth:** which edges light up
- **135°:** mounds look like pits
- **z factor:** exaggerates small relief
- **Multidirectional:** all orientations at once

DGM1, DOM1: © Geobasis NRW, dl-de/zero-2-0

Hillshade: renderer or new raster

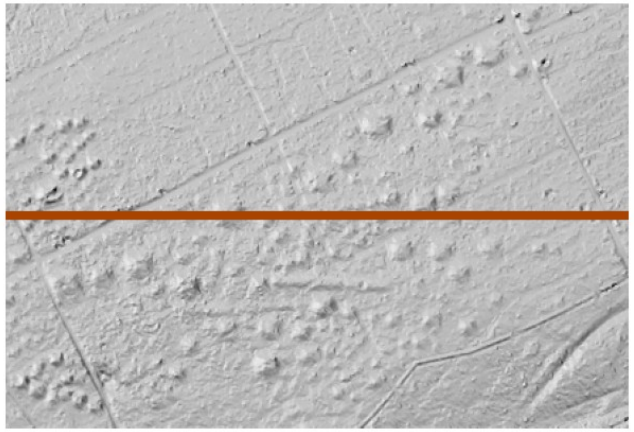
- **Renderer:** display only, try settings live
- **Hillshade tool:** a new raster 0–255 for layouts and analysis
- Same method (Horn, 3 × 3 window)

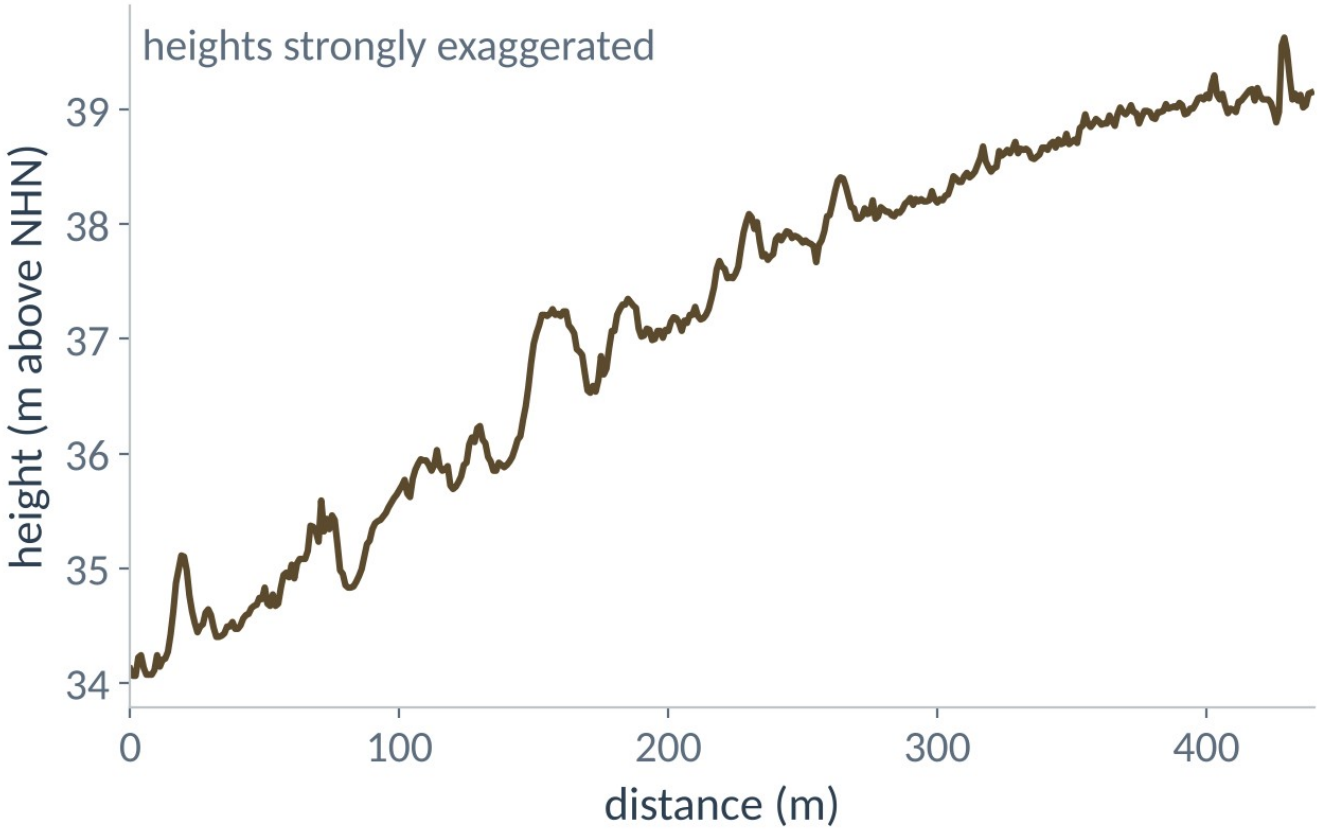
```
from osgeo import gdal

gdal.DEMProcessing(
    "hillshade.tif", "uedem.vrt",
    "hillshade", zFactor=2,
    multiDirectional=True)
# slope in degrees:
gdal.DEMProcessing(
    "slope.tif", "uedem.vrt", "slope")
```




A profile across the mounds





QGIS: View → Elevation Profile · layer: Elevation → Represents Elevation Surface

DGM1, DOM1: © Geobasis NRW, dl-de/zero-2-0

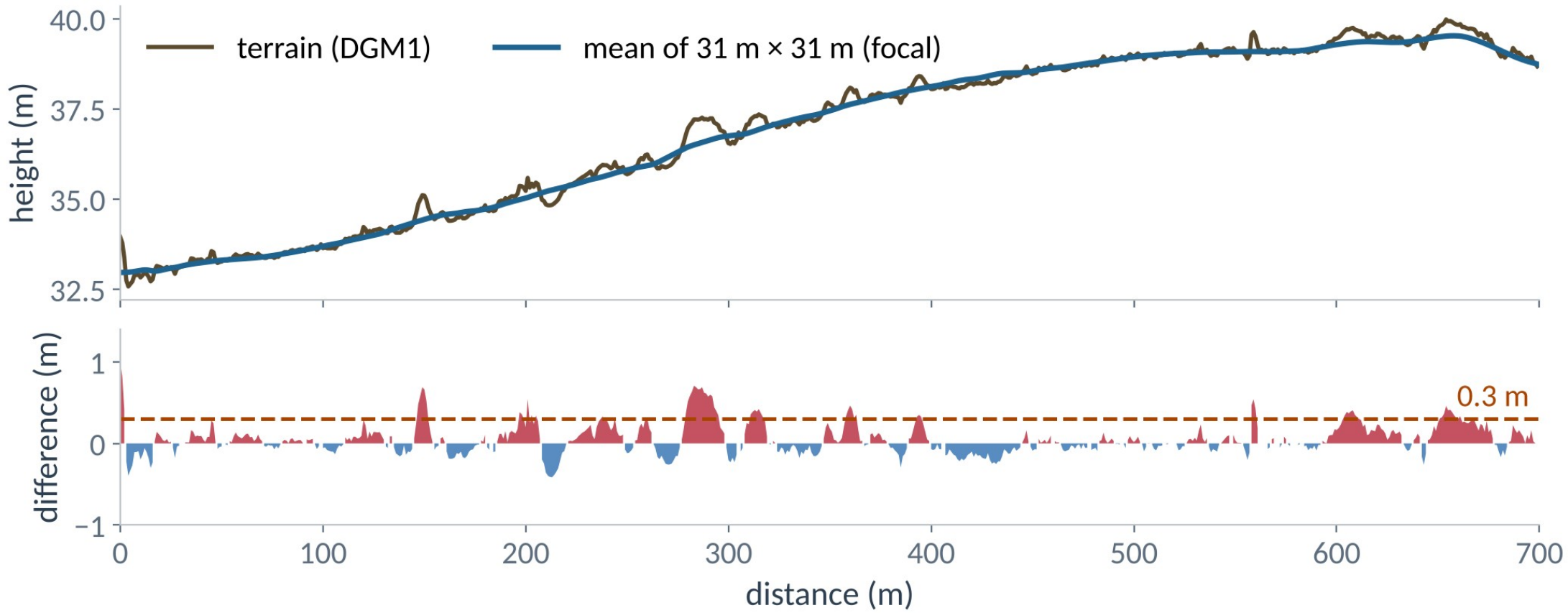
4 Local relief

Map algebra on the terrain

Terrain models | Tiles | Hillshade | **Local relief** | Mounds



Local relief model: terrain minus smoothed terrain



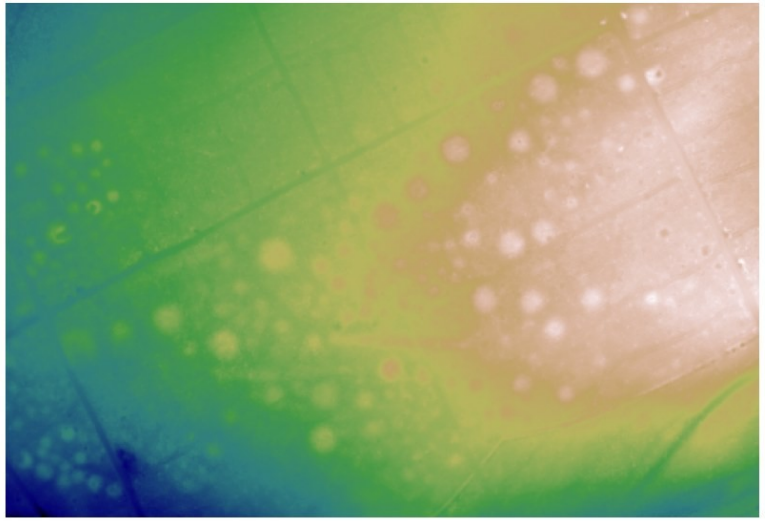
focal (moving mean) → **local** (difference) → **local** (threshold 0.3 m)

Wikipedia: Moving average · High-pass filter

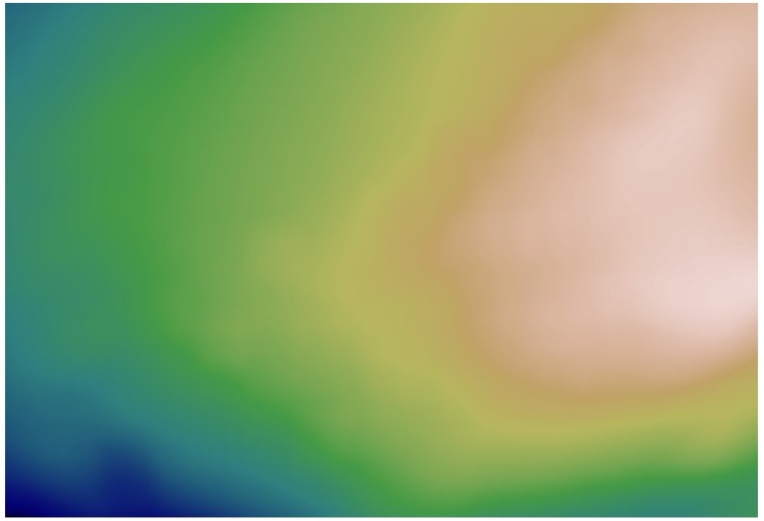
DGM1, DOM1: © Geobasis NRW, dl-de/zero-2-0 · R. Hesse (2010), Archaeological Prospection 17, 67–72

Local relief in three steps

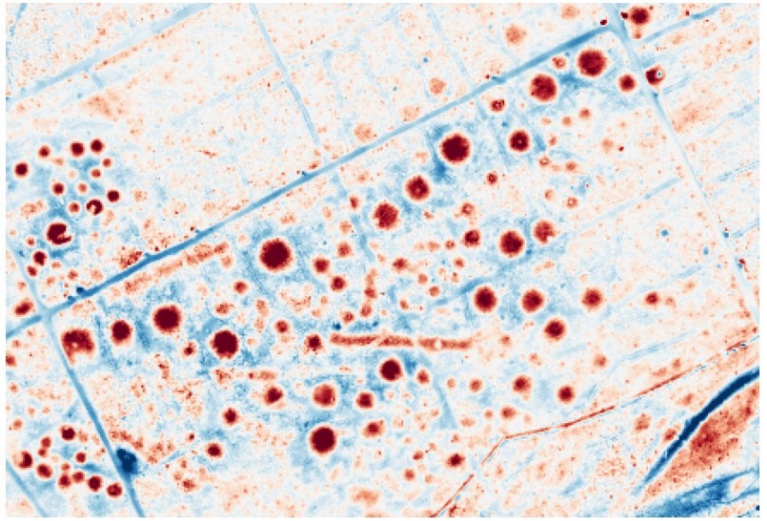
terrain



– smoothed (31 m)



= local relief



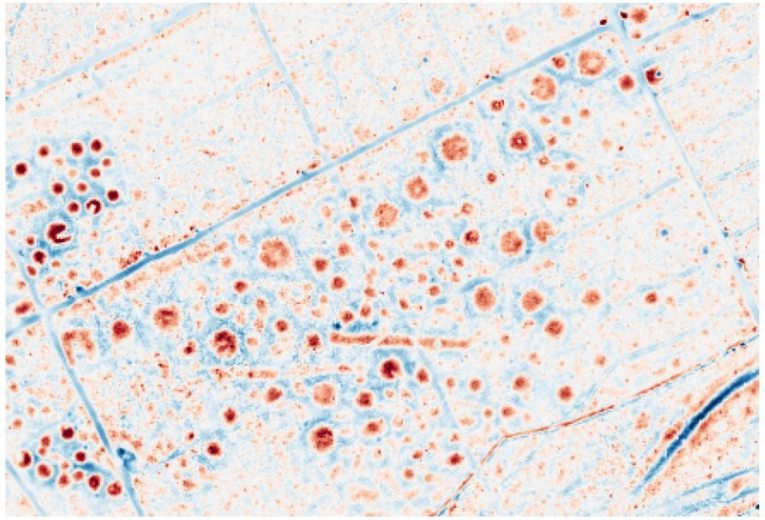
```
# focal: mean of 31 x 31 cells
mean = scipy.ndimage.uniform_filter(dtm, 31)
# local: difference and threshold
lrm = dtm - mean
mask = lrm > 0.3
```

DGM1, DOM1: © Geobasis NRW, dl-de/zero-2-0

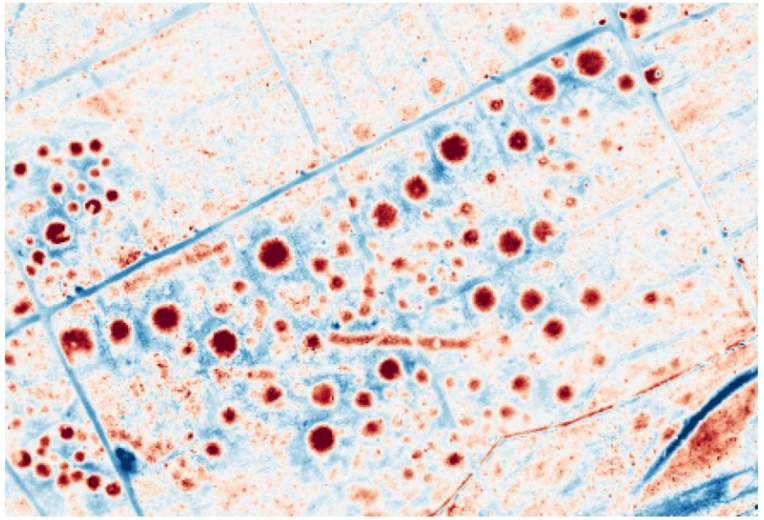


The window decides what you see

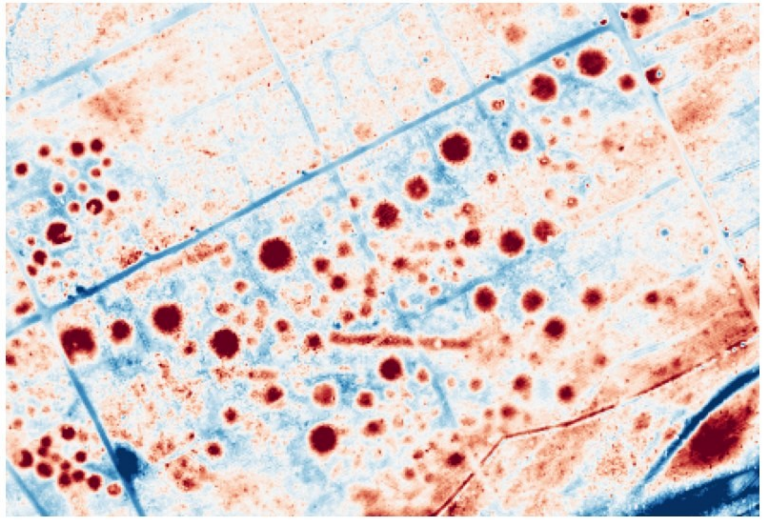
window 15 m



window 31 m



window 61 m

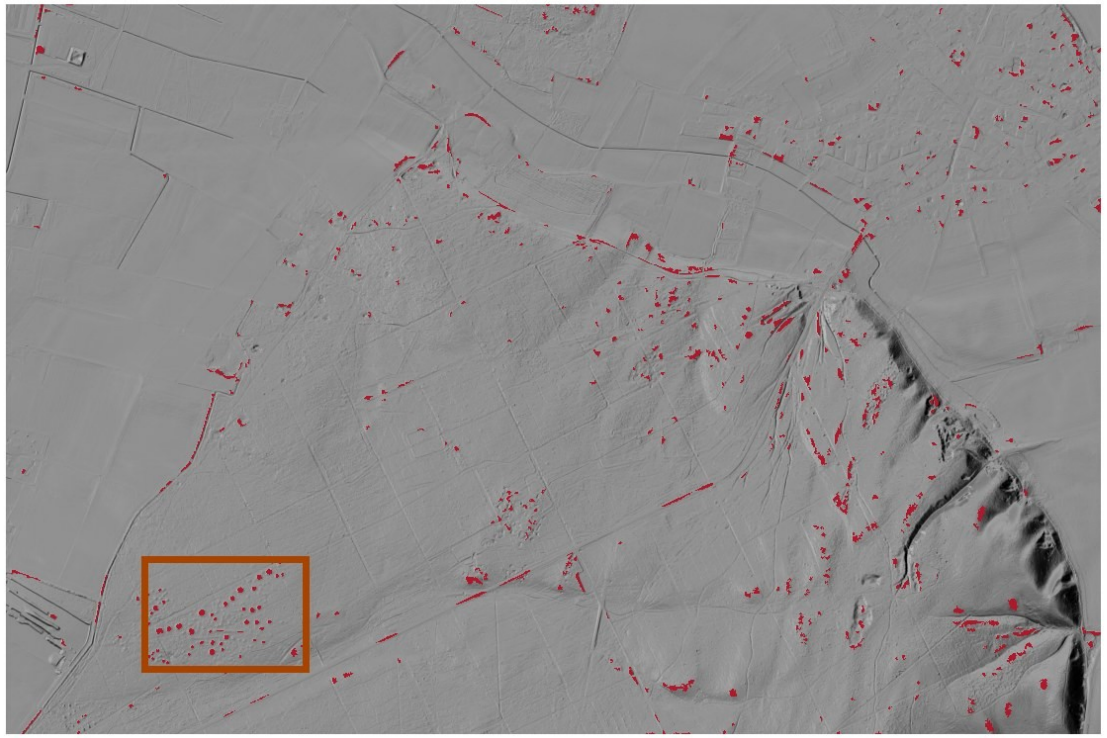


Rule of thumb: window ≥ 2 × the diameter of what you look for, but not much larger

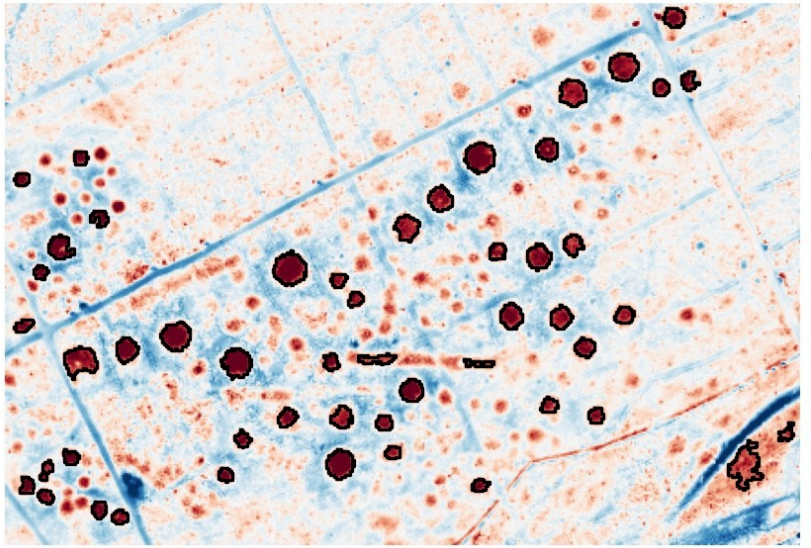
DGM1, DOM1: © Geobasis NRW, dl-de/zero-2-0

From bumps to candidates

565 candidates in the six tiles



51 in the mound field



Pause: 565 candidates, about 50 mounds: what else is a bump?

DGM1, DOM1: © Geobasis NRW, dl-de/zero-2-0

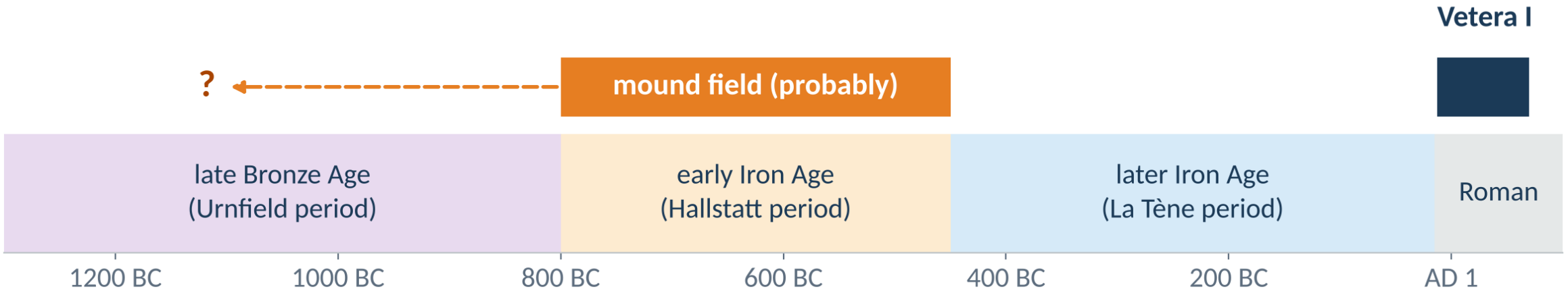
5 Mounds

Map, terrain model and algorithm

Terrain models | Tiles | Hillshade | Local relief | **Mounds**



Who built the mounds?



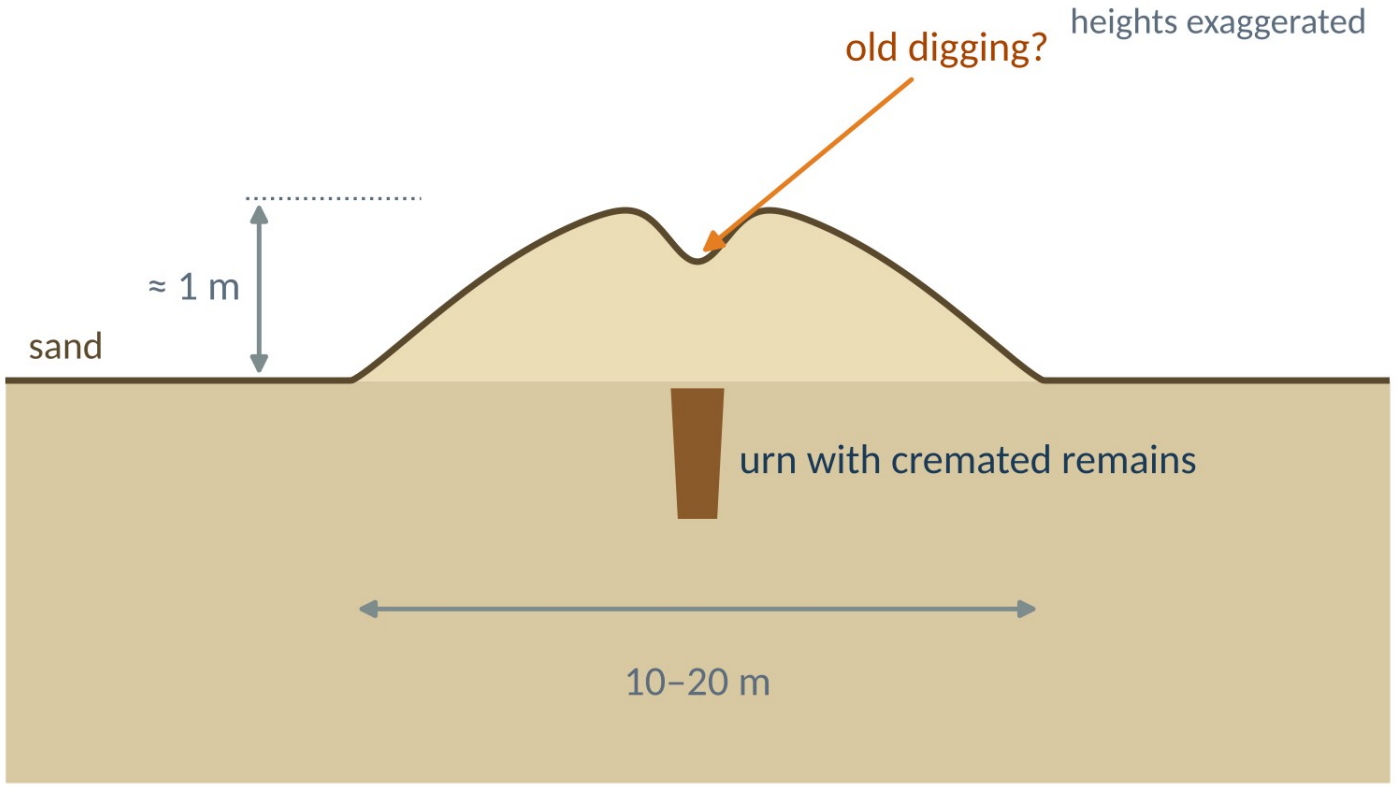
- **Who:** farming communities of the Lower Rhine; no name, no village found
- **When:** probably early Iron Age, about 800–450 BC (Hallstatt period)
- **How:** cremation; the ashes in an urn under a low mound of dune sand
- **How many:** > 40 visible, perhaps > 200 graves (Lower Rhine tumulus culture)

Wikipedia: Uedemer Hochwald (de) · Hallstatt period · Urnfield culture

Preliminary, to be confirmed with the LVR-Amt für Bodendenkmalpflege · Wikipedia: Uedemer Hochwald (CC BY-SA)

geo1040 · DTM tiles, hillshade, local relief · © 2026 Rolf Becker · EOLab-HSRW · CC BY-SA 4.0

What is in a mound?



- Cremation, urn in the centre
- Covered with dune sand, ≈ 1 m high today
- Pit in the top: old diggings?
- Survived under forest, ploughed flat in fields

Wikipedia: Tumulus · Cremation

A protected monument

- **Ground monument** (Bodendenkmal): marked K.D. on the old map
- No digging, no metal detectors, no collecting without a permit
- **Look, measure, map:** the terrain model lets us study every mound without touching it
- Found something? Report it to the LVR-Amt für Bodendenkmalpflege (Xanten)

Wikipedia: Bodendenkmal (de)

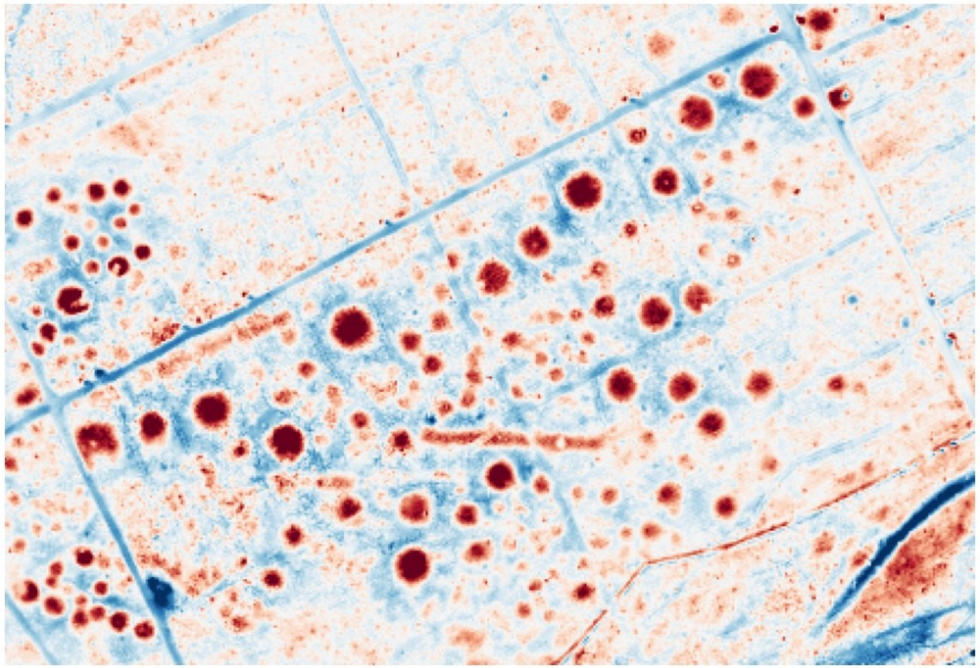


Burial mounds

older map: mounds as dots (no coordinates)



local relief of the DGM1

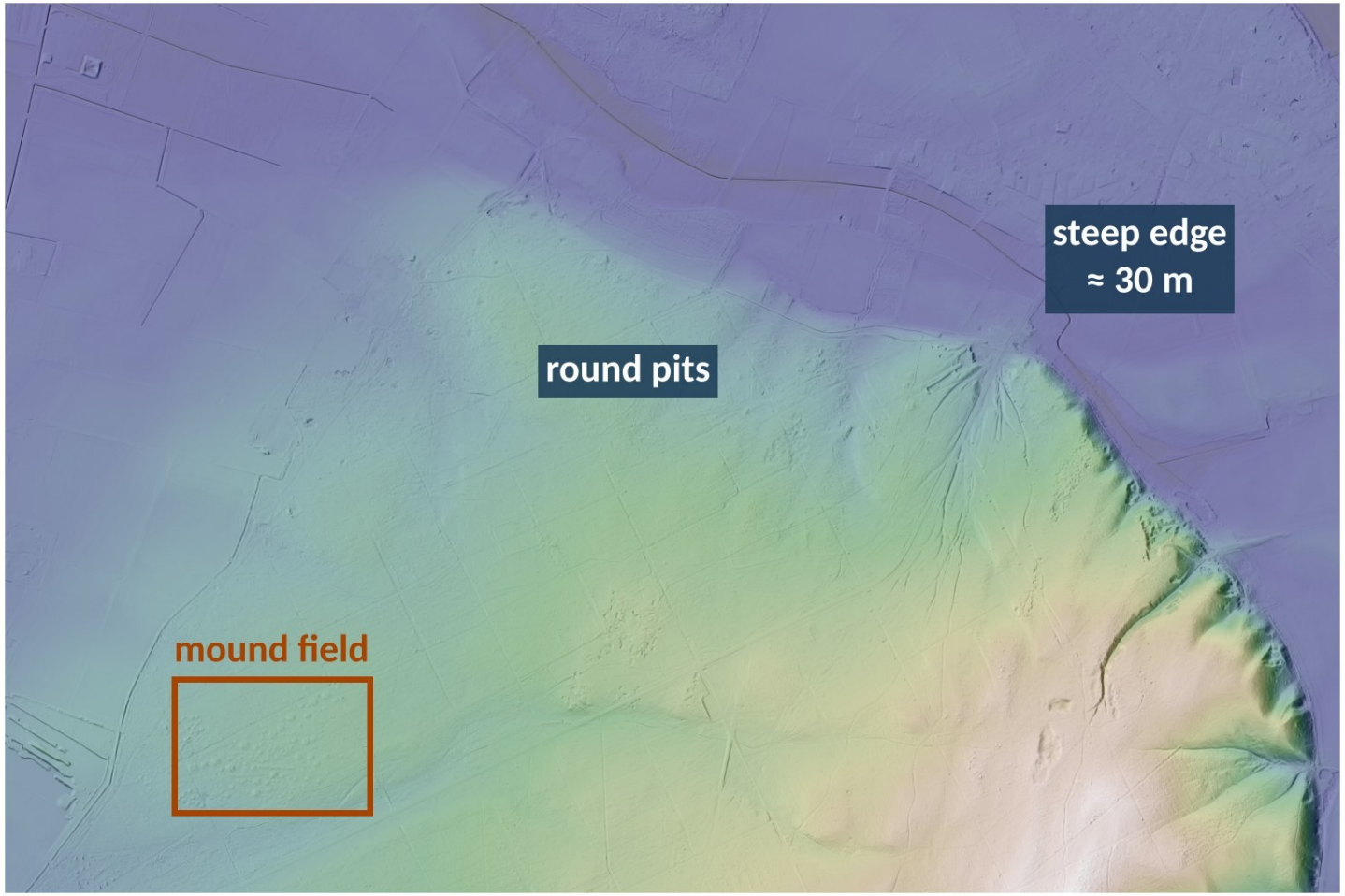


Homework geo1042: georeference the map, digitise the mounds from the terrain model, compare map ↔ terrain ↔ algorithm (51 candidates in the field)

Wikipedia: Tumulus · Hügelgrab (de)

Map extract: © Geobasis NRW, dl-de/zero-2-0 · DGM1, DOM1: © Geobasis NRW, dl-de/zero-2-0

Why is there a steep edge in the forest?

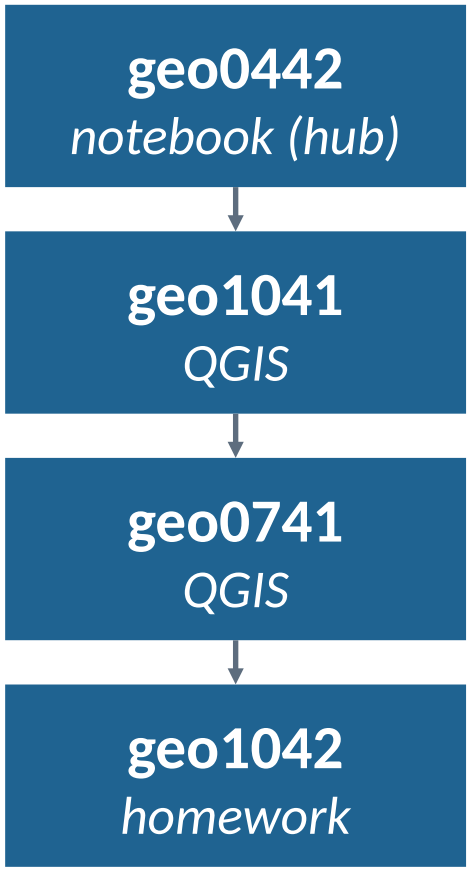


- Push moraine of the Saale glaciation
- Edge ≈ 30 m, cut by gullies
- Round pits: negative forms, ≈ 0.6 m deep
- **Your job:** describe, measure, then interpret

Wikipedia: Push moraine · Saale glaciation · Niederrheinische Höhen (de)

DGM1, DOM1: © Geobasis NRW, dl-de/zero-2-0

Today: four activities, one chain



tile list → index → select → download → zip

virtual raster → hillshade → profile

smooth → subtract → threshold → polygons

georeference map → digitise mounds → compare

Summary

- A terrain model is the ground; a surface model includes trees and roofs
- Tiles are found by name and joined without copying: a virtual raster
- A hillshade makes small relief visible; it is not a height
- Local relief = terrain − smoothed terrain: focal, then local map algebra
- **The window size decides what you find**



Sources and further reading

Laser scanning from the air: Lidar · Airborne laser scanning (de)

Surface model or terrain model?: Digital elevation model · Digitales Geländemodell (de)

One layer from six files: the virtual raster: Tiled web map · GDAL

Hillshade: light from one direction: Terrain cartography · Hillshading (de)

Local relief model: terrain minus smoothed terrain: Moving average · High-pass filter

Who built the mounds?: Uedemer Hochwald (de) · Hallstatt period · Urnfield culture

What is in a mound?: Tumulus · Cremation

A protected monument: Bodendenkmal (de)

Burial mounds: Tumulus · Hügelgrab (de)

Why is there a steep edge in the forest?: Push moraine · Saale glaciation · Niederrheinische Höhen (de)

Text distilled from Wikipedia (CC BY-SA 4.0), the Geobasis NRW product information and R. Hesse (2010). Data: DGM1, DOM1, DOP, map extract © Geobasis NRW, dl-de/zero-2-0.
Figures: R. Becker (make_slides.py), CC BY-SA 4.0.



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- **Contact:** EOLab-HSRW, Hochschule Rhein-Waal, Kamp-Lintfort · eolab.de